

CO5	-	-	2	-	2	-	-	-	2	-	-	2	-	-	-
Average	-	-	0.4	-	2	-	-	-	0.6	-	-	0.4	-	-	-

1 – Weakly Mapped (Low)

2 – Moderately Mapped (Medium)

3 – Strongly Mapped (High)

“-” means there is no correlation

Syllabus

Unit I: Introduction to Backend Development,

12 Lecture Hours

Server-Side Programming

Overview of backend development and its importance, Client-server architecture and communication protocols, Introduction to HTTP and RESTful APIs, Introduction to backend languages and frameworks (e.g., Python, Node.js, Django, Flask), Introduction to server-side programming languages (e.g., Python, JavaScript), Basics of handling HTTP requests and responses, Managing server-side sessions and cookies, Introduction to server-side rendering and template engines.

Unit II: Data Management, API Development

12 Lecture Hours

and Frameworks

Introduction to databases (relational and NoSQL), Designing database schemas and models, Performing CRUD (Create, Read, Update, Delete) operations on databases, Query optimization and indexing, designing, and implementing RESTful APIs, Authentication and authorization mechanisms, Handling API requests and responses, API versioning and documentation.

Unit III: Frameworks and Performance Optimization

12 Lecture Hours

Introduction to popular backend frameworks (e.g., Express.js, Django, Ruby on Rails), Working with middleware and routing, integrating third-party APIs and services, Testing and debugging backend applications, understanding security vulnerabilities and best practices, Implementing secure authentication and authorization mechanisms, Caching and performance optimization techniques, Error handling and logging.

Unit IV: Deployment and DevOps

12 Lecture Hours

Overview of deployment environments (e.g., local, cloud, containers), Monitoring and scaling backend applications, Basics of containerization (e.g., Docker), Continuous integration and deployment (CI/CD), Managing infrastructure configurations, setting up CI/CD pipelines, Building and packaging applications, Automated testing and code quality checks, Continuous Deployment and Release Management, Strategies for releasing software updates, Managing deployment environments, Release orchestration and rollback strategies.

Unit V: Advanced Topics

12 Lecture Hours

Real-time communication with Web-Sockets, Microservices architecture and design patterns: Overview of microservices architecture and its benefits, Service discovery and service registration, Implementing communication between microservices (REST, messaging), Serverless computing: Understanding serverless architecture and its advantages, Building serverless backend systems using platforms like AWS Lambda or Google Cloud Functions, Event-driven serverless architecture and integration with other services, Function-as-a-Service (FaaS), Performance profiling and optimization strategies.

Total lecture Hours 60

References*

Textbooks	1. David Herron, "Node.js Web Development" 5th Edition, Packt Publication, 2020. 2. DT Editorial Services, "HTML5 Black Book", 2nd Edition, Dreamtech Publications, 2016. 3. Sam Newman, "Building Microservices", 2nd Edition, O'Reilly Publication, 2021.
Reference books	
Web Resources	
Journals	
MOOCs, online courses	

Modes of Evaluation: Quiz/Assignment/ presentation/ extempore/ Written Examination etc.

Examination Scheme

Components	IA	MID SEM	End Sem	Total
Weightage (%)	50	20	30	100

Backend Development Lab

List of Experiments

- Experiment1** Create a web page with all possible elements of HTML5
- Experiment2** Create a web page with all types of Cascading style sheets
- Experiment3** Create a Responsive Web page with HTML and CSS
- Experiment4** Create Responsive web page with Bootstrap
- Experiment5** Programs to demonstrate JavaScript array, object and functions
- Experiment6** Client-Side Scripts for Form Validation using JavaScript
- Experiment7** Programs to familiarise ES6 concepts
- Experiment8** Programs to demonstrate DOM and event handling.
- Experiment9** Programs using AJAX with HTML, XML and JSON data
- Experiment10** Programs to familiarize Query.(2 program)
- Experiment11** Create a website with HTML, CSS and JavaScript (implement Ajax)
- Experiment12** Programs to familiarize Server-Side Scripting using Node JS
- Experiment13** Programs using Mongo DB database with Node JS

Total Lab hours 30

References*

Textbooks	1. David Herron, "Node.js Web Developent", 5th Edition, Packt Publication, 2020. 2. DT Editorial Services "HTML5 Black Book", 2nd Edition, Dreamtech Publications, 2016. 3. Sam Newman, "Building Microservices", 2nd Edition, O'Reilly Publication, 2021.
Reference books	
Web Resources	
Journals	

MOOCs, online courses	
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Modes of Evaluation: Quiz/Assignment/ presentation/ extempore/ Written Examination etc.

Examination Scheme: Continuous Assessment

Components	Quiz & Viva	Performance & Lab Report
Weightage (%)	50 %	50 %